FROM A POLYNOMIAL RELATION WITH COEFFICIENTS IN R, THE LEADING COEFFICIENT TIMES T IS INTEGRAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Integrality from a monic polynomial relation. Definition of integrality, applied at the end of the counted proof (line 30712). Turns the rewritten relation into a monic equation for the element a_n t over R.
- Multiplying a polynomial relation by a power of the leading coefficient. The computation performed in the counted proof at lines 30706-30712. Rescales the relation so that its top term becomes the n-th power of a_n t.

Lemma 0.1. *Let $\varphi : R \to S$ be a ring map. Suppose $t \in S$ satisfies the relation $\varphi(a_0) + \varphi(a_1)t + \ldots + \varphi(a_n)t^n = 0$. Then $\varphi(a_n)t$ is integral over R.*

Proof. Namely, multiply the equation $\varphi(a_0) + \varphi(a_1)t + \ldots + \varphi(a_n)t^n = 0$ with $\varphi(a_n)^{n-1}$ and write it as $\varphi(a_0a_n^{n-1}) + \varphi(a_1a_n^{n-2})(\varphi(a_n)t) + \ldots + (\varphi(a_n)t)^n = 0$. □

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